

MolPallette — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallette/coconut-flavordb_v4/naveja_recap

generated 2026-08-20 12:29

A • Provenance

field	value
sources	['flavordb', 'coconut']
method	naveja_recap
decomposition_params	{'ratio': 0.3333333333333333, 'include_ring': True, 'max_cores':
core_ratio	0.3333333333333333
max_cores	4
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	4412
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	3
rdkit_version	2026.03.2
created_at	2026-08-20T12:28:07

B · Composition

quantity	value
molecules	413,458
from coconut	391,964 (94.8%)
from flavordb	21,494 (5.2%)
decompositions	1,329,619
per molecule	mean 3.22, median 4, max 4
R-group slots	1,792,185
per decomposition	mean 1.35
k >= 2	26.74% of decompositions

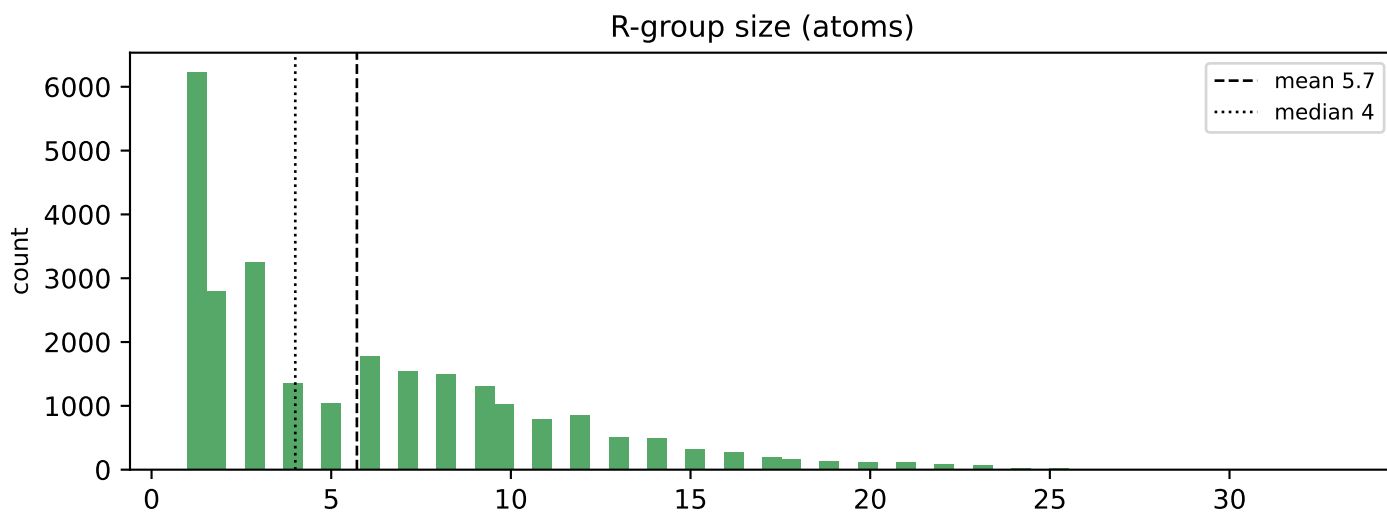
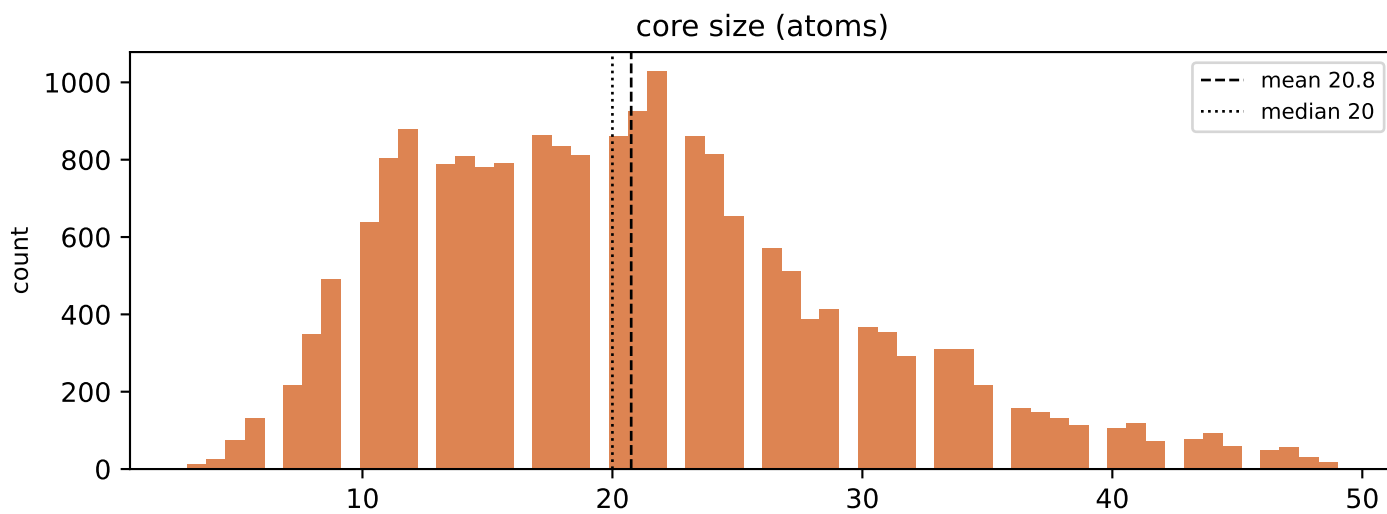
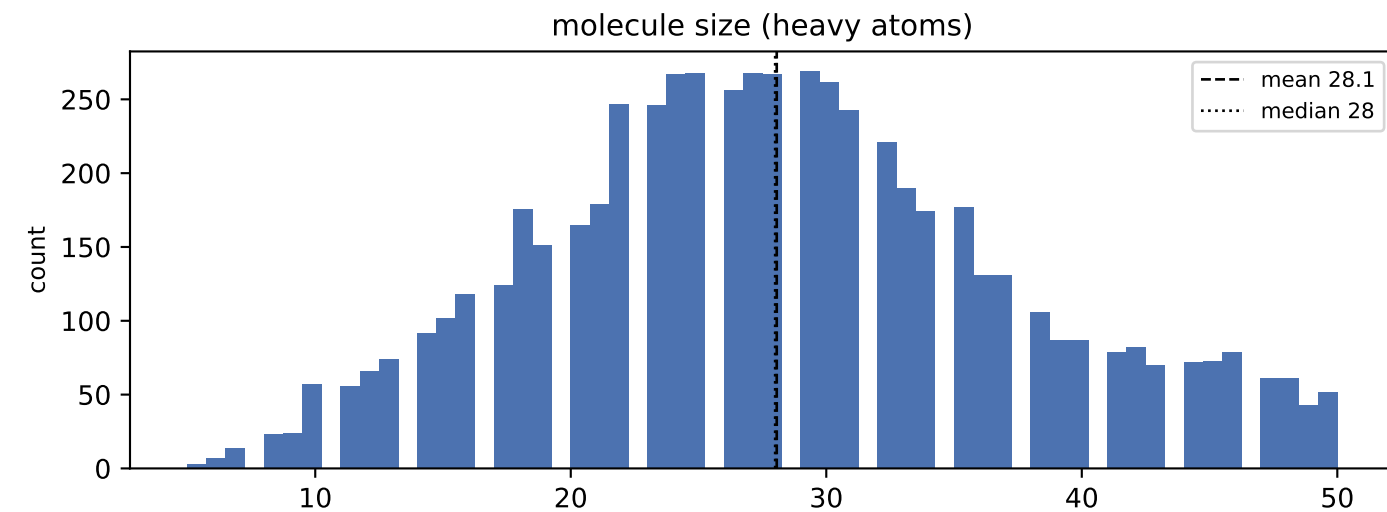
k>=2 is what the islinked subset space and the core-decoration objective have to work with.

C · Augmentation space

scheme	size
sampling_unit = molecule	413,458 instances/epoch
sampling_unit = decomposition	1,329,619 instances/epoch (3.2x)
full (mol, core, islinked) space	2,692,335 (6.5 per molecule)

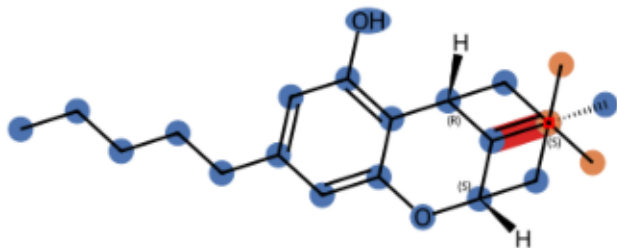
MolPLA's two stages: one molecule yields many cores; one core yields $2^k - 1$ islinked subsets.

D • Size distributions

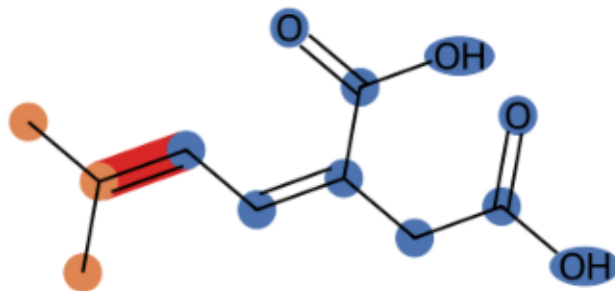


E · Example decompositions (ratio 0.3333333333333333)

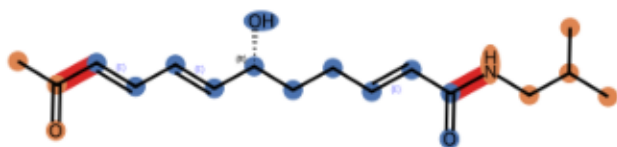
CNP0000074 k=1 core 20 atoms, R-groups [3]



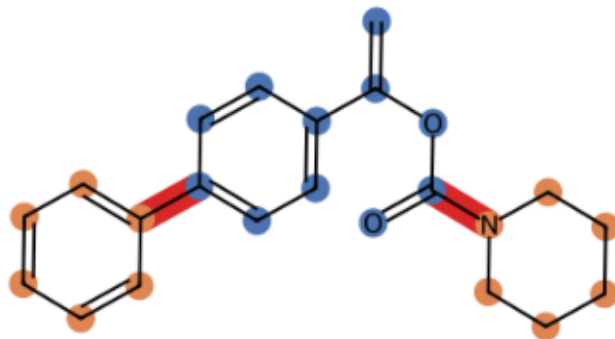
CNP0000161 k=1 core 10 atoms, R-groups [3]



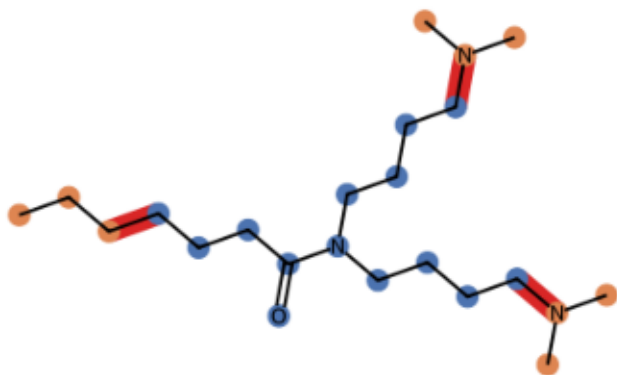
CNP0000307 k=2 core 12 atoms, R-groups [3, 5]



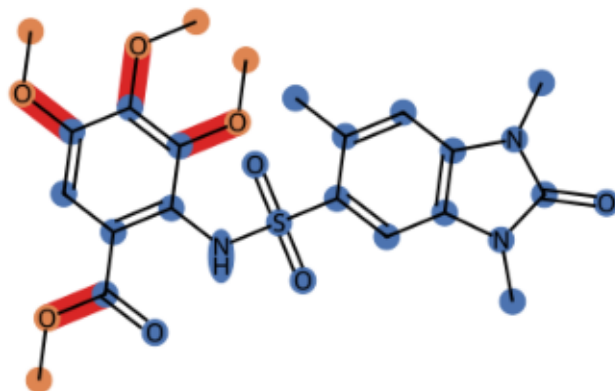
CNP0001292 k=2 core 11 atoms, R-groups [6, 6]



CNP0003527 k=3 core 14 atoms, R-groups [3, 3, 3]

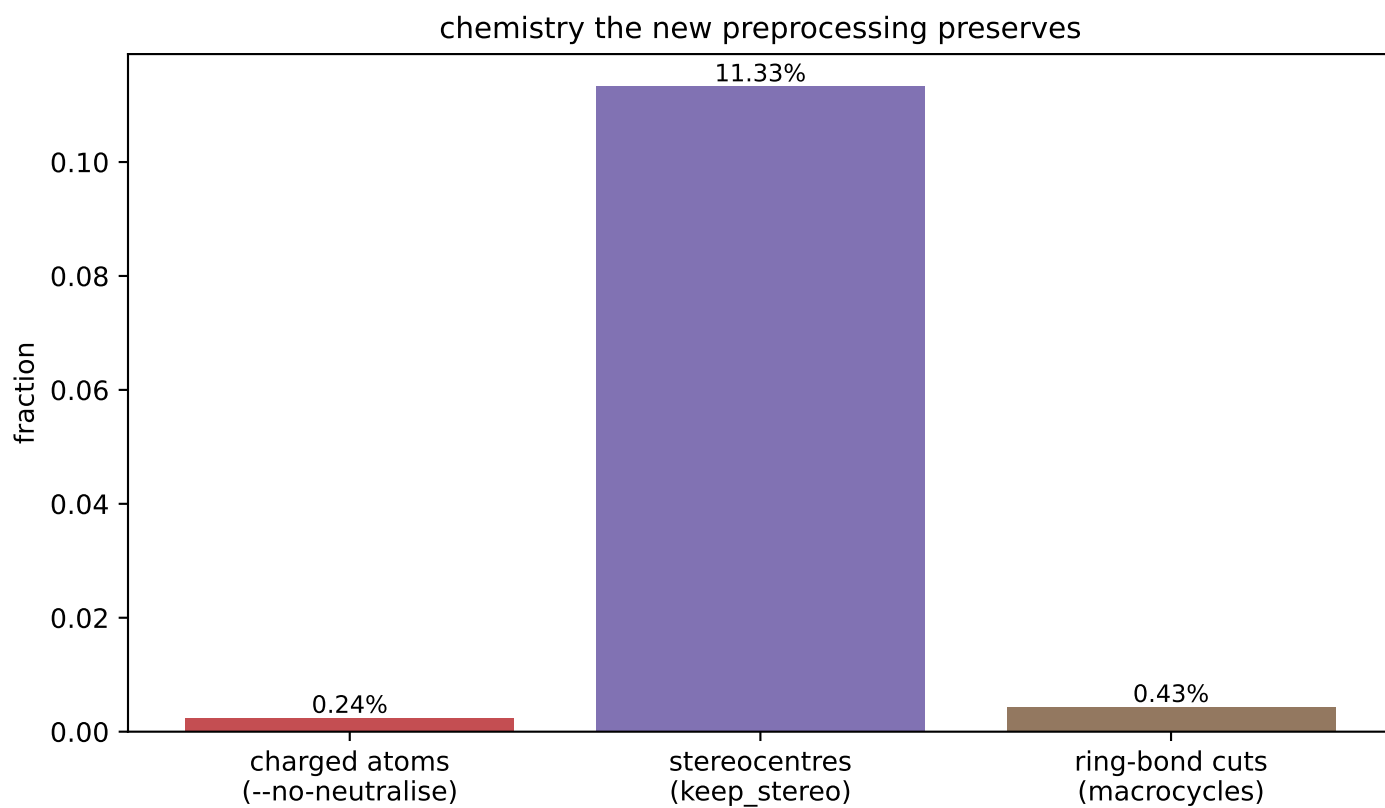
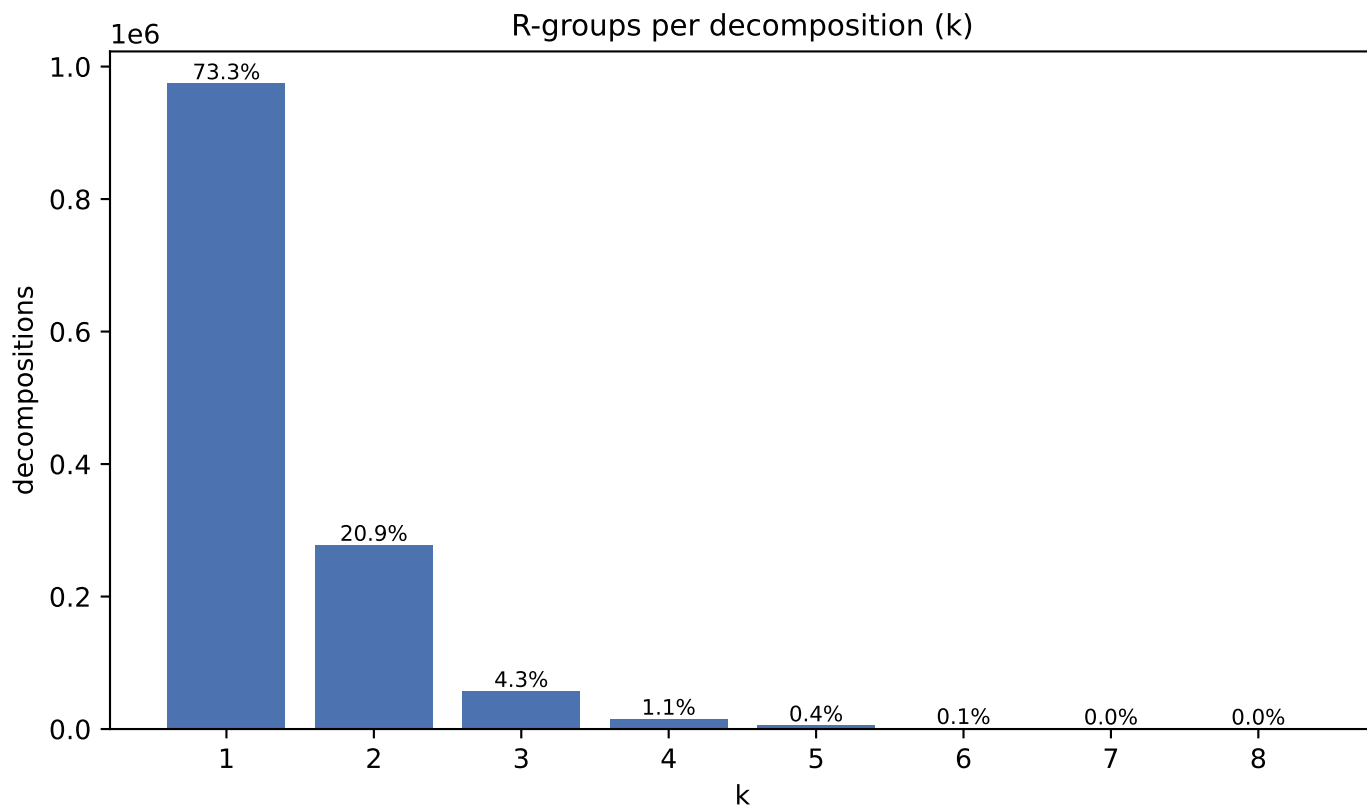


CNP0009431 k=4 core 25 atoms, R-groups [2, 2, 2, 2]

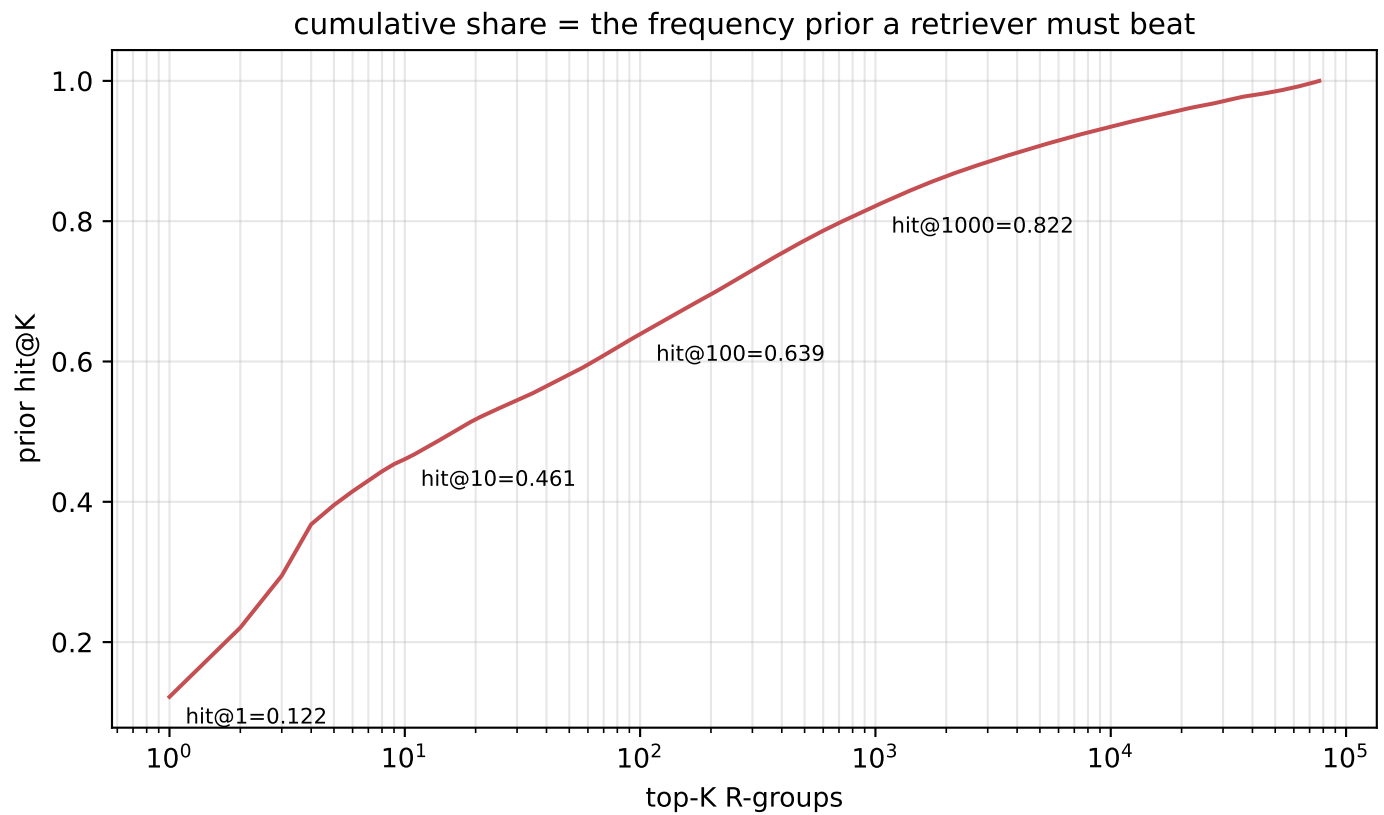
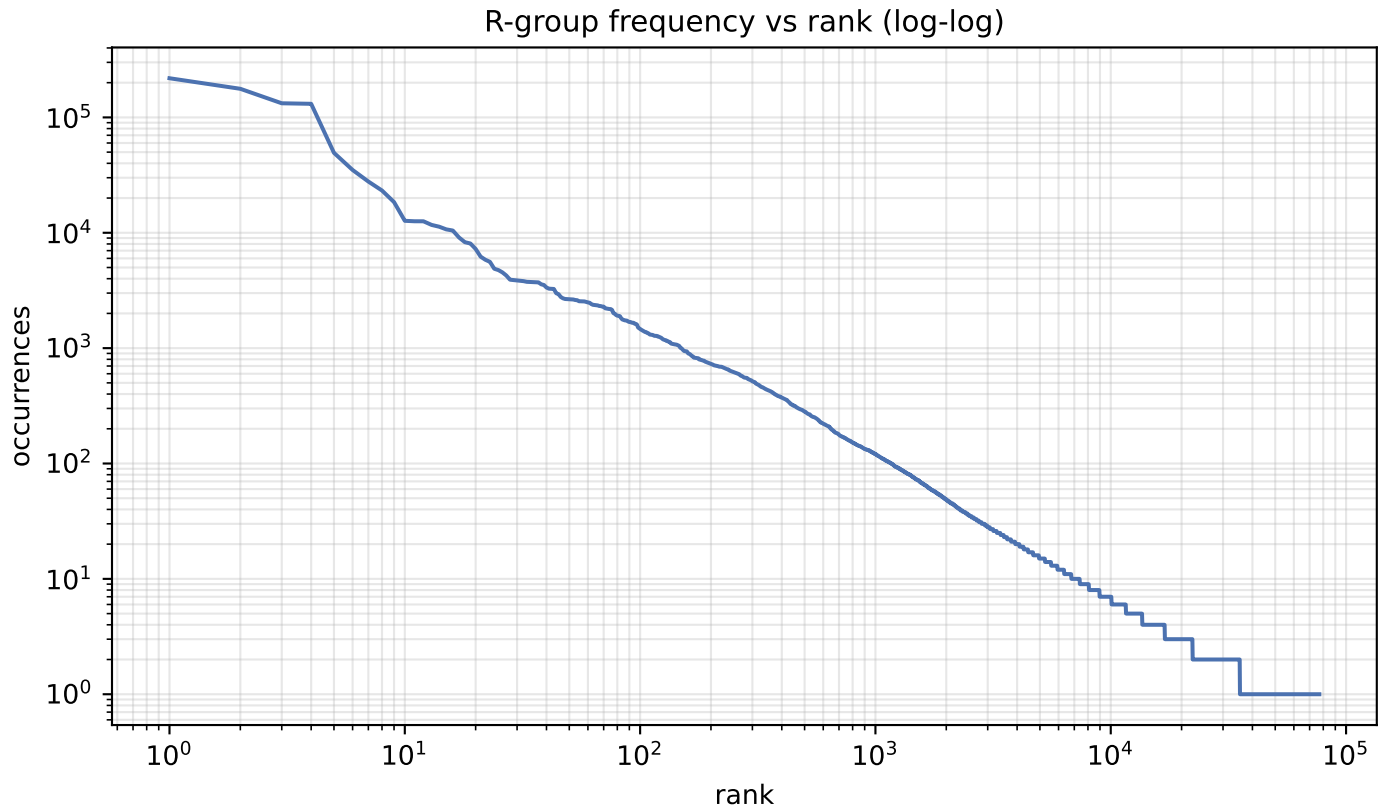


blue = core orange = R-groups red = cut bonds

F · Decomposition shape & preserved chemistry



G · R-group library skew



H · R-group library

quantity	value
distinct R-groups	76,961
occurrences	1,792,185
effective size (exp H)	385
top-1 share	12.19%
top-10 cover	46.1%
top-100 cover	63.9%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
218,534	12.19%	*C
176,996	9.88%	*O
132,555	7.40%	*OC
131,185	7.32%	*C(C)=O
49,310	2.75%	*C(C)C
35,022	1.95%	*c1ccccc1
27,804	1.55%	*CC
23,237	1.30%	*OC(C)=O
18,447	1.03%	*Cc1ccccc1
12,698	0.71%	*O[C@@H]1O[C@H](CO)[C@H](O)[C@H](O)[C@H]1O
12,582	0.70%	*C(=O)c1ccccc1
12,575	0.70%	*C(=O)O
11,697	0.65%	*CO
11,286	0.63%	*c1ccc(OC)cc1
10,704	0.60%	*OCC
10,452	0.58%	*N
9,098	0.51%	*CC(=O)O
8,287	0.46%	*c1ccc(O)cc1
8,071	0.45%	*CCC
7,241	0.40%	*CCO

J · Instance-level common-R-group filter

quantity	value
percentile	99.9
count threshold	1,992
hashes flagged common	77 of 76,961
their share of occurrences	61.7%
single-R-group instances rejected	15,884/26,048 (61.0%)
mean R-group size, rejected	2.94 atoms
mean R-group size, kept	10.05 atoms

Filters INSTANCES, not the library: the retrieval target space and its frequency prior are unchanged. At k=1 the rule reduces to 'drop if the R-group is common', so rejection is far heavier than MolPLA saw.